



ya diggg: Advanced Technical Spec & Logic Analysis

This document addresses the search logic (Semantic vs Fuzzy) and the feasibility of audio previews for the ya diggg companion.

1. Search Logic: "The Hybrid Semantic" Model

The user asked about **Chroma (Vector)** vs **Fuzzy Matching**. For a low-power ARM Cortex-A53 device, a pure on-device semantic engine is too heavy.

Approach	On-Device Logic	Pro	Con
Fuzzy (Current)	SQLite LIKE & Jaro-Winkler	Ultra-low power; fast for text.	Fails on noisy OCR or concept shifts.
Pure Semantic	Local VectorDB (Chroma)	High "concept" intelligence	Heavy RAM use; slow on-device embedding TryChromaReddit .
Proposed: Hybrid	sqlite-vss (Faiss)	Best of both worlds.	Requires pre-computing on Desktop.

Decision: Hybrid Semantic Search.

- **Desktop App:** During USB-C sync, the desktop app generates vector embeddings for the record database.
- **On-Device:** ya diggg performs a lightweight **K-Nearest Neighbor (KNN)** search using the pre-computed vectors. This keeps the device fast and cool while offering "Chroma-level" intelligence

2. Audio Preview: "The 50,000 Song Library"

Feasibility analysis for adding audio previews to the 32GB ya diggg.

A. Storage Calculation

- **Preview Length:** 30 Seconds.
- **Bitrate:** 128kbps (MP3/OGG).
- **File Size:** ~0.48 MB per preview [Optiviewusa](#)
- **Capacity:** On 25GB usable eMMC, ya diggg can store **~52,000 individual track previews**.

B. Hardware Architecture

Component	Choice	Rationale
Audio DAC	ESS ES9218PC	Hi-Res capable, ultra-low power consumption for handhelds Hifiwalker
Jack	3.5mm Analog	Maintain the "Discreet Instrument" DNA. Zero latency, no software pairing bugs.
Wireless	Murata 1DX (Optional)	Combo Wi-Fi/BT chip. Use ONLY if user wants wireless headphones support

3. The "Sync-to-Play" Workflow (Zero Streaming)

To maintain the "Offline Soul," ya diggg does not stream from Spotify/Apple.

1. **Sync:** When connected via USB-C, the user's desktop client fetches 30s previews for their "Wishlist" or "Digging List."
 2. **Storage:** Previews are indexed to the local SQLite database.
 3. **Interaction:** User scans a record -> Reveals Facts -> **Long-press Orange Button** to hear the 30s preview through the 3.5mm jack.
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4. Updated Scope & Risks

- **Risk:** On-device vector search for 1M+ records may exceed 512MB RAM.
- **Mitigation:** Use a **Tiered Index**. Full-text search (FTS5) for primary Artist/Title, and Vector Search (sqlite-vss) for "Recommended lineage" or noisy archival labels.
- **Decision on Wi-Fi:** We will **provisionally include a Wi-Fi/BT module** on the PCB but keep it **firmware-disabled** by default to preserve battery, enabling it only for optional "Wireless Headphone" or "Cloud Sync" modes.